

MODEL FORMULATION & ASSUMPTIONS

# The Linear Model, Revisited

*A simplified trend, four assumptions, and what happens when real data breaks them*

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# By the end of this lecture, you'll be able to...

- 1 Explain what a linear regression model actually claims about a real-world trend
- 2 State the four assumptions behind linear regression (LINE), and what each one protects
- 3 Recognize when real data violates these assumptions, using a diagnostic plot
- 4 *If we have time*: Identify which extension — transformation, GLM, or mixed model — addresses which violation

# The model you already know

$$Y_i = a + bX_i + \varepsilon_i$$

## Independent & dependent variables

$X$  = age.  $Y$  = height.

$$\hat{y}_i = \hat{a} + \hat{b}x_i$$

The least-squares estimate of  $E[Y_i | x_i]$ .

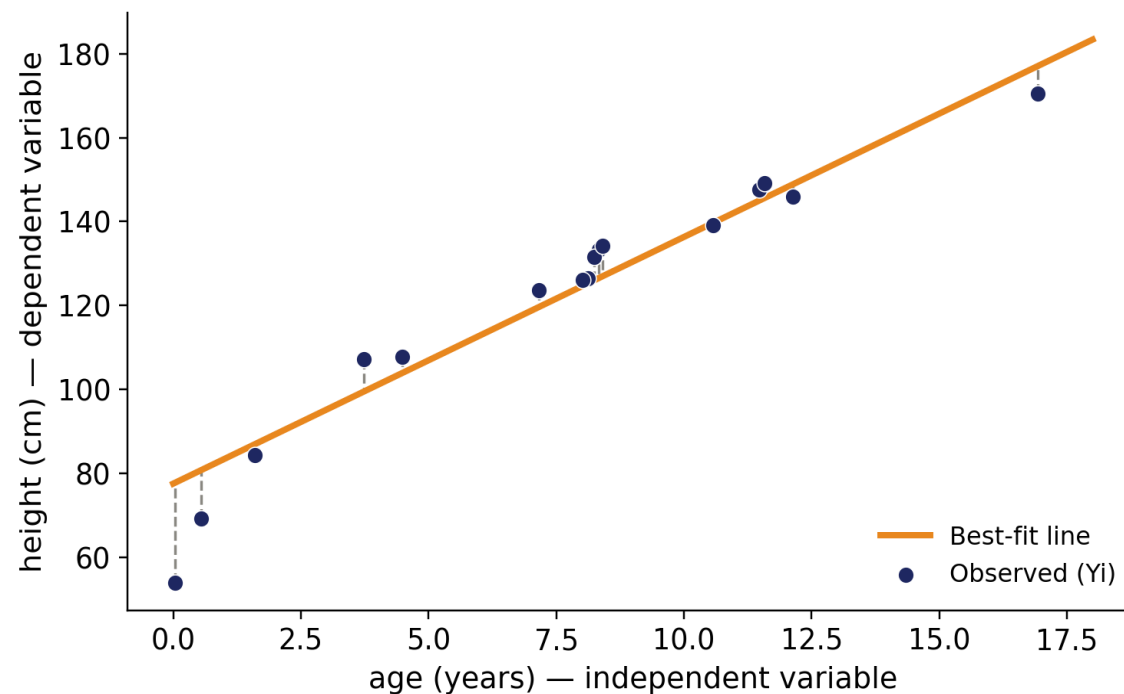
$\varepsilon_i$

True error term (“noise” or unexplained variation). We assume  $E[\varepsilon_i] = 0$ .

$$e_i = Y_i - \hat{y}_i$$

The residual (our sample estimate of the error). Its mean is 0 by construction due to the least-squares fitting

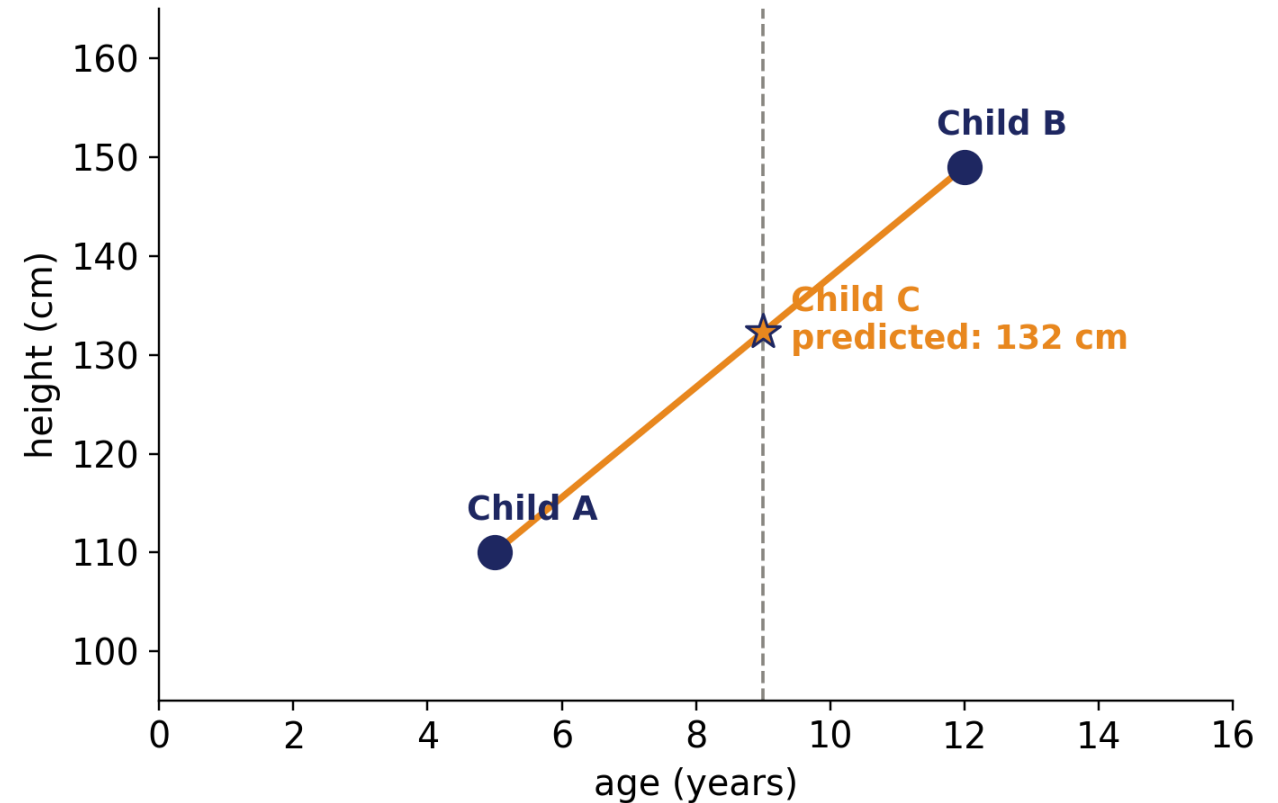
*“All models are wrong, but some are useful” — Box (1976)*



*A model is a simplified trend: the part we describe, plus everything ( $\varepsilon$ ) we don't.*

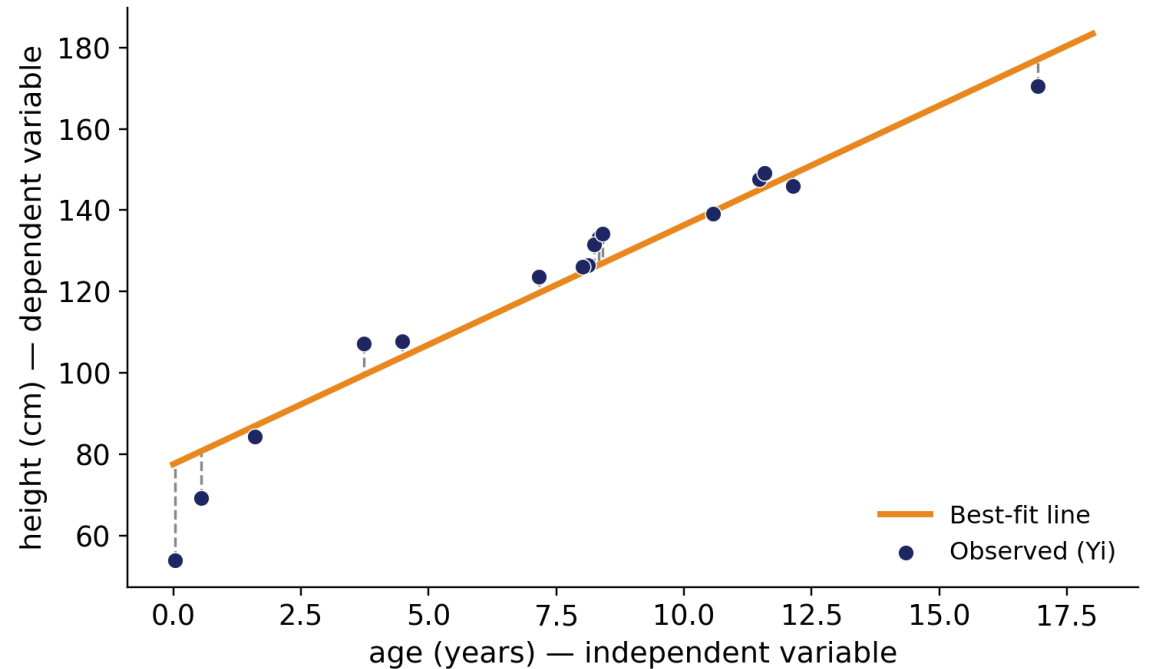
# Predict Child C's height

- Two data points:
  - Child A: age 5, height 110 cm
  - Child B: age 12, height 149 cm
- Child C is 9. Connect A and B, read off the value
- Predicted height: about 132 cm. This is the regression you already know



# Scaling up to a whole class

- Two children was enough to connect a line — a whole class needs a best-fit line instead
- Same idea as Child A and B, just with more points pulling on the line
- Every dashed line is one child's residual, the same  $e_i$  from the recap slide, now visible for a whole class
- A good fit means residuals are small and randomly scattered, without any systematic pattern (which will get back to later)



# LINE, precisely

*A model is a simplified trend. LINE is the four assumptions linear regression relies on — checked by examining the raw relationship between the variables, the fitted parameters, and how the residuals behave.*



## Linear

The functional form is correctly specified: the model is linear in its parameters, and a linear relationship holds between the independent variable(s) and  $E[Y]$ .



## Independent

The observations and their residuals are independent of one another. Knowing one tells you nothing about another.



## Normal

The residuals are normally distributed: most errors cluster near zero, with progressively fewer large ones, forming a symmetric bell curve.



## Equal variance

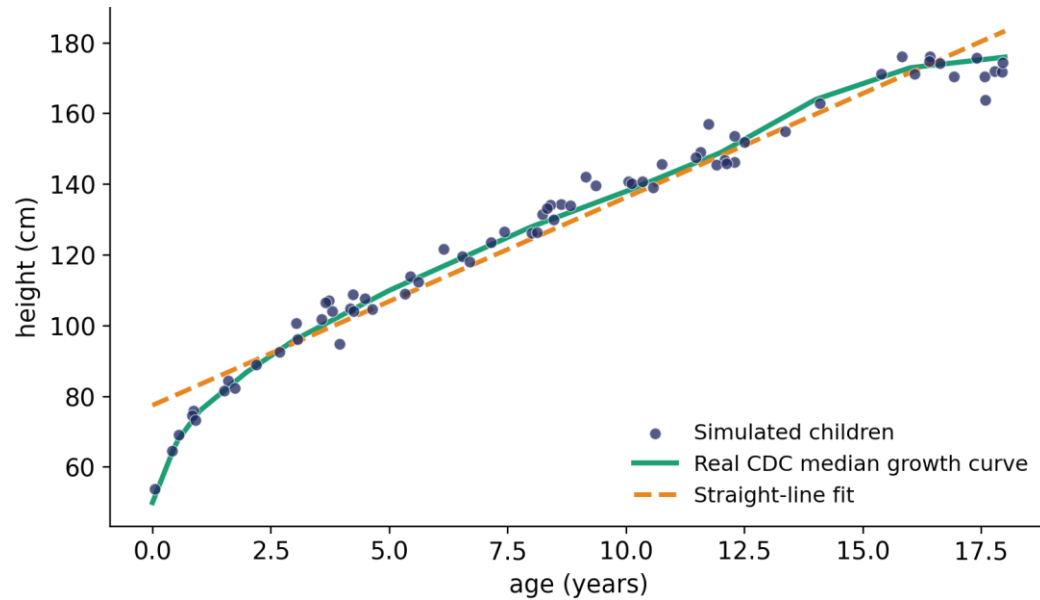
The spread of the residuals stays constant across every value of the independent variable; the noise does not grow or shrink.

## Now let's put LINE to the test

- So far, one independent variable: age (one independent variable)
- **HOWEVER**, real growth depends on genetics, sex, nutrition, illness, timing of puberty — age is one piece of a much bigger picture
- From here, we will also track weight alongside height, which is another window into the same growth process
- A model that only sees age is going to leave a lot on the table — and LINE is exactly where that shows up

# L — Linear

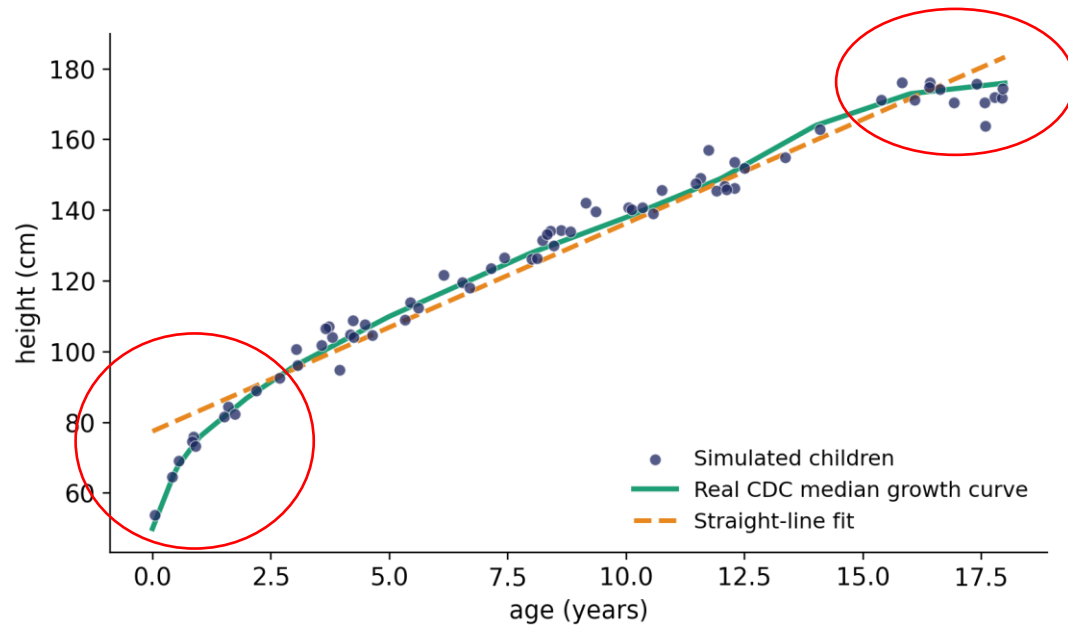
Remember the best-fit line from earlier?



*Real CDC median growth curve (green) vs. a straight-line fit (orange) through simulated children.*



# L — Linear

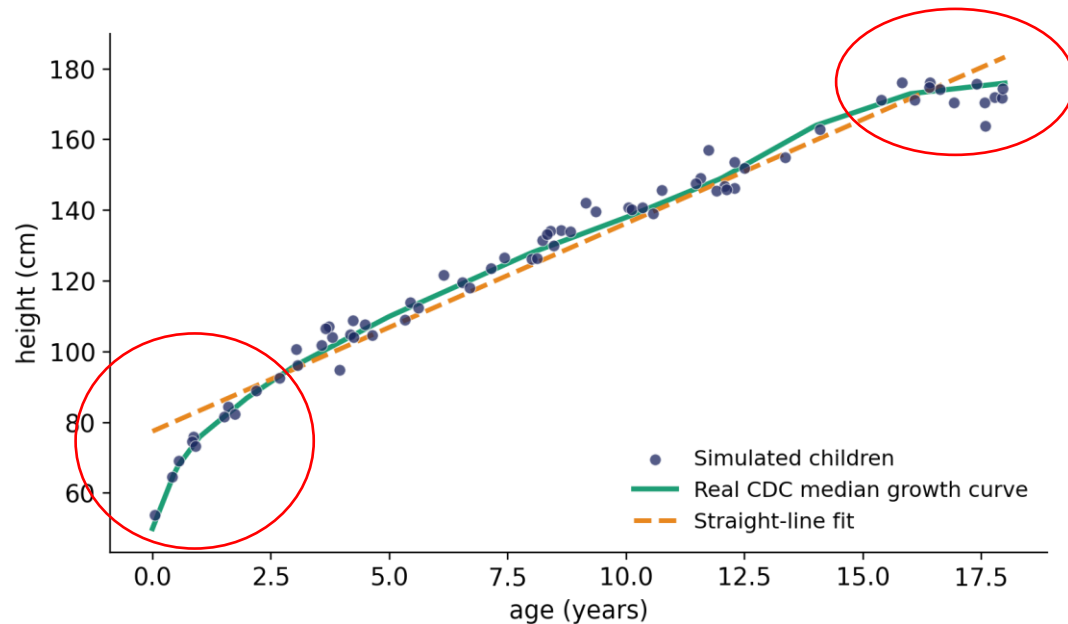


Here is what it was missing

*Real CDC median growth curve (green) vs. a straight-line fit (orange) through simulated children.*

# L — Linear

Real growth is fast in infancy, steadier through mid-childhood, then a pubertal spurt and a plateau

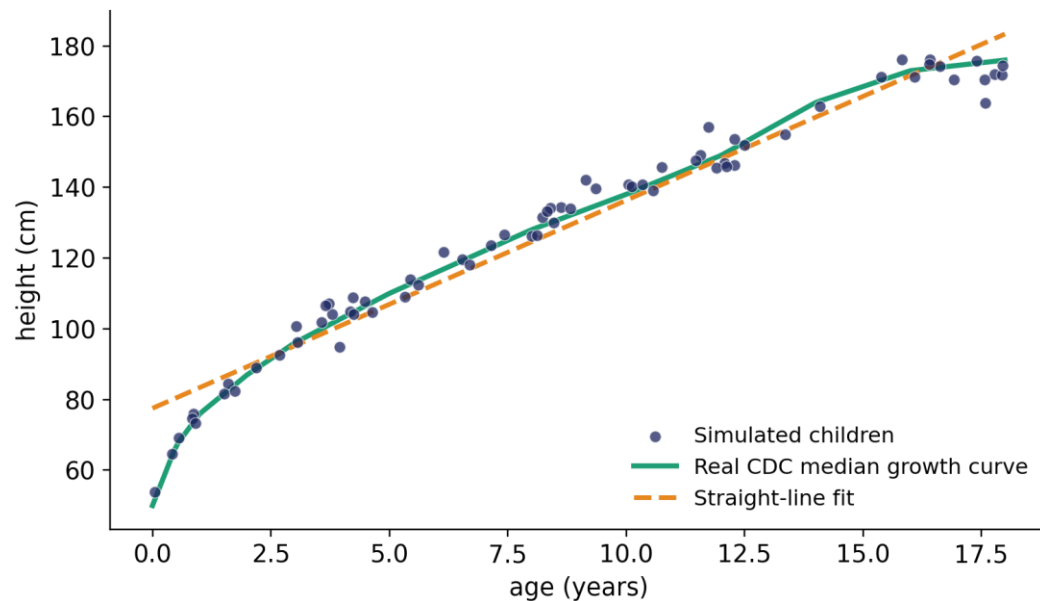


The straight-line fit badly misses both ends

*Real CDC median growth curve (green) vs. a straight-line fit (orange) through simulated children.*

# L — Linear

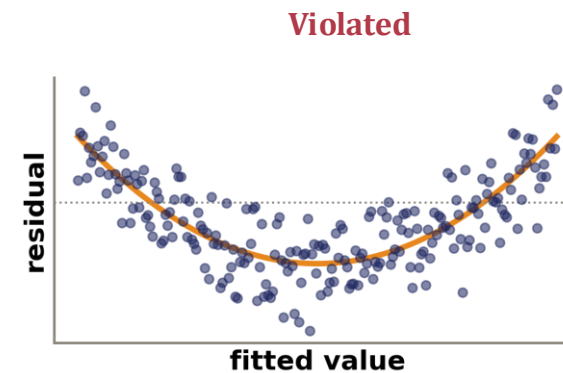
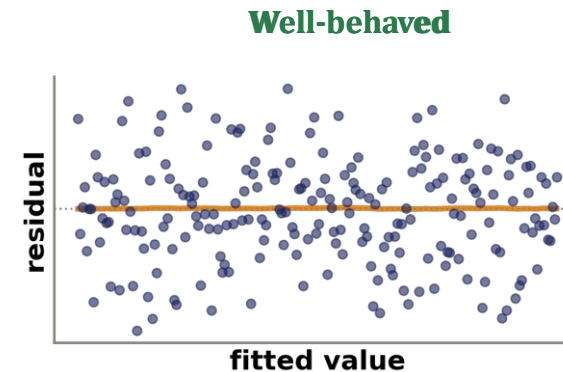
Real growth is fast in infancy, steadier through mid-childhood, then a pubertal spurt and a plateau — *a straight line misses both ends*



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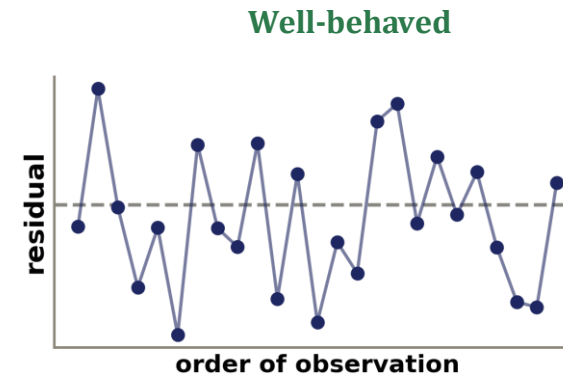
Formal test: Ramsey RESET.

Fixed by: richer X (more terms, still linear in the parameters) — covered next.



# I — Independent

Once we know their ages, Child A's residual tells us nothing about Child B's.

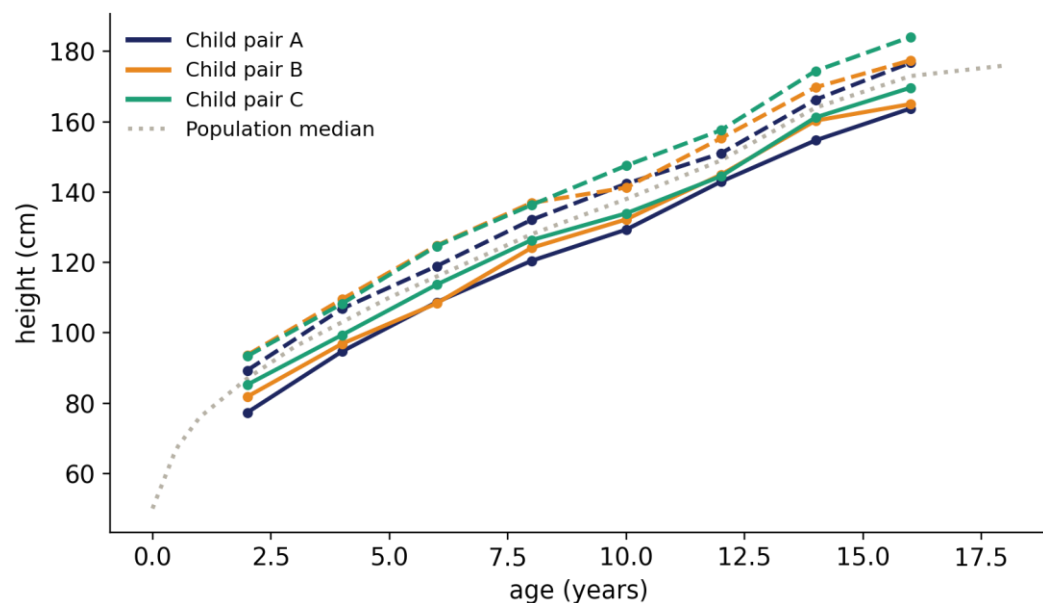


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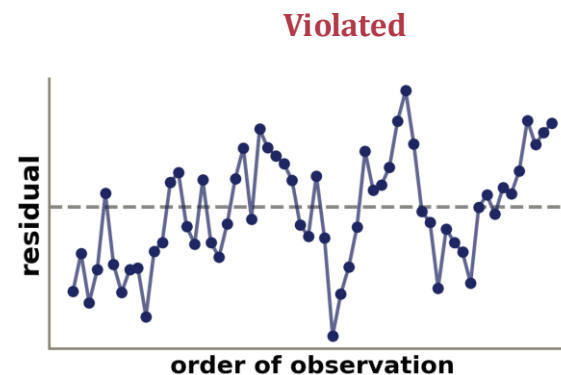
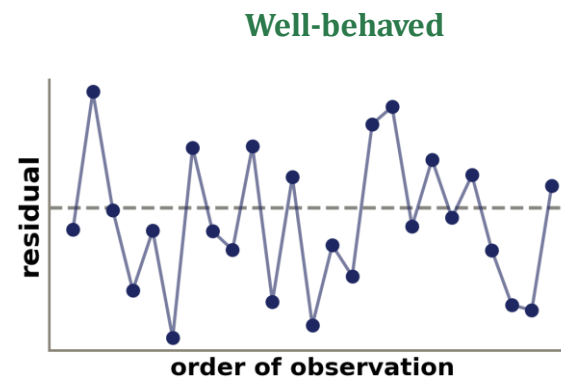
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Repeated measurements on the same child violate this;

Two siblings are being measured within the dataset: given that height is heritable trait, these two measurement are not independent



Three pairs of simulated children, each measured repeatedly across the same ages.

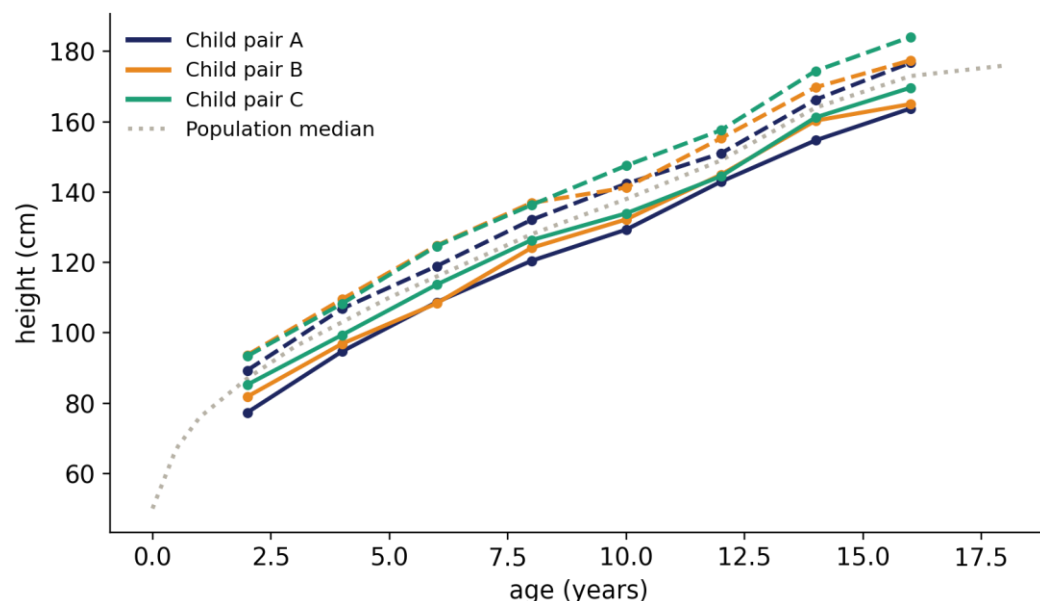


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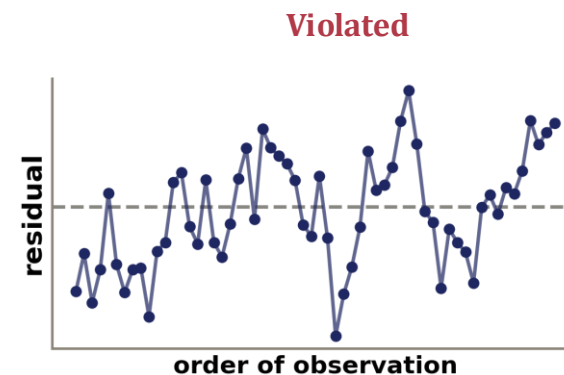
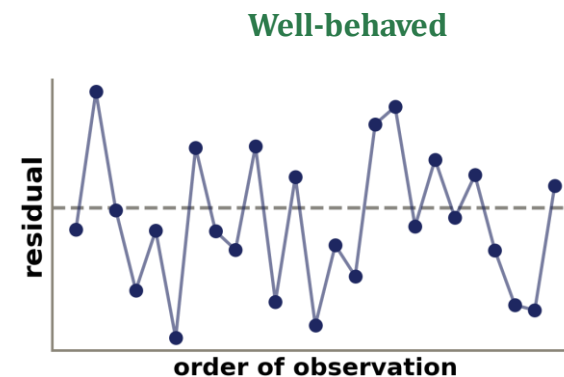
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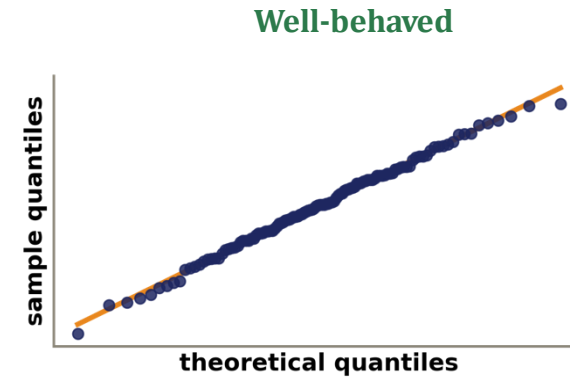
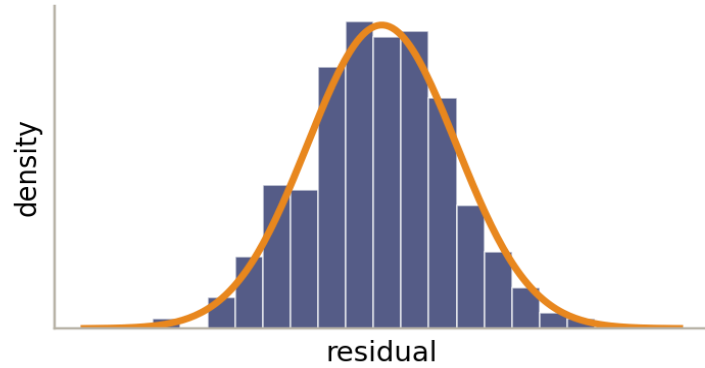
Formal test: Durbin-Watson.

Fixed by: mixed models (random effects absorb the correlation) — covered next.



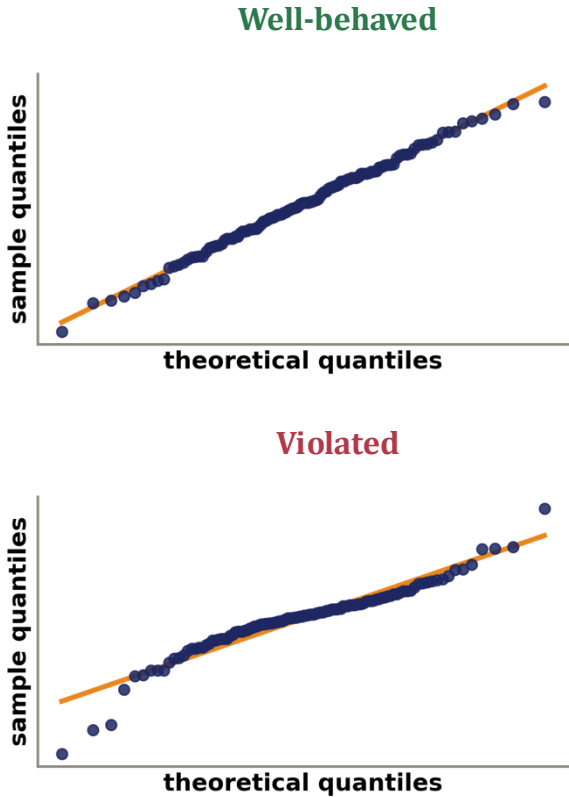
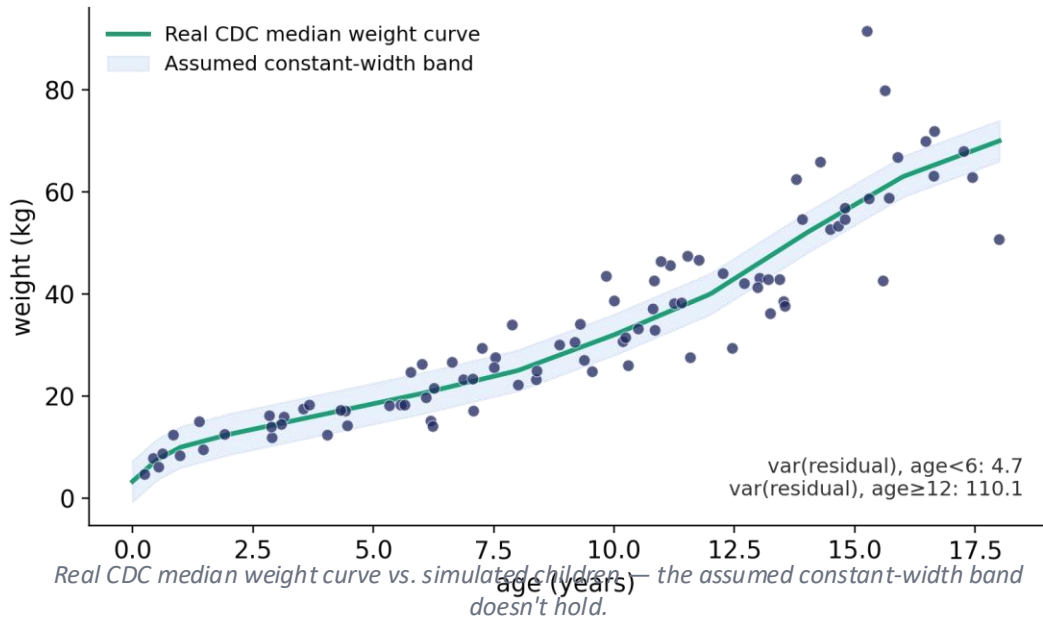
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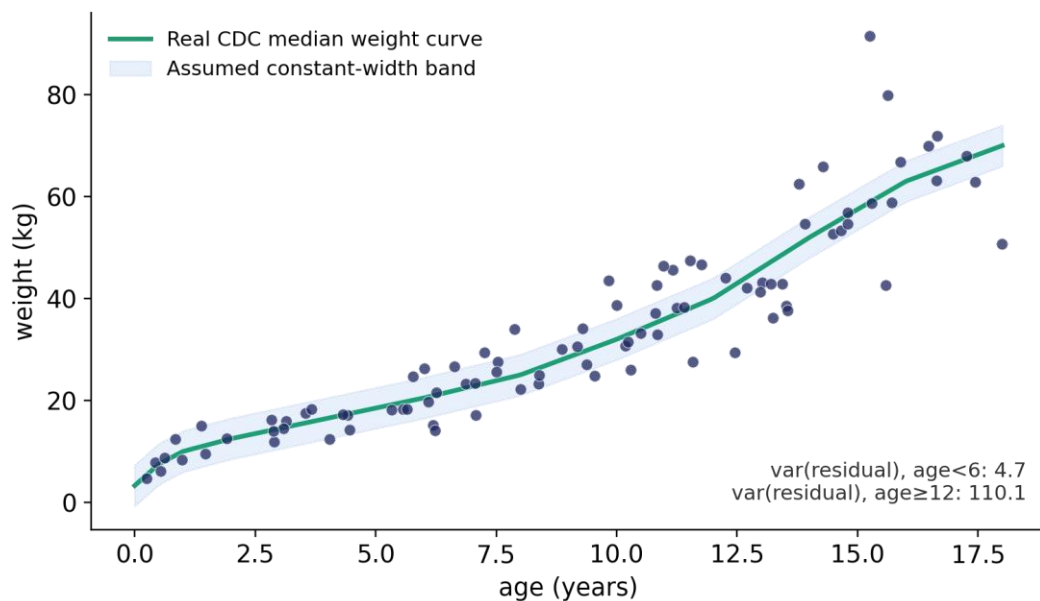




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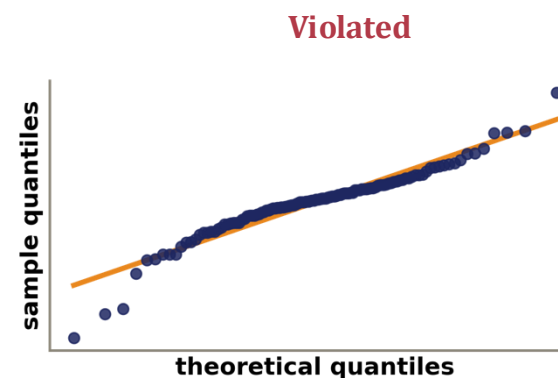
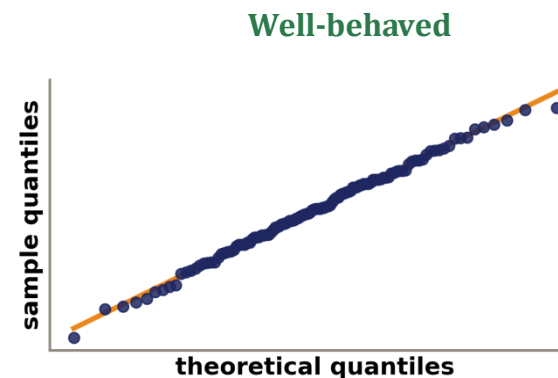


*Real CDC median weight curve vs. simulated children — the assumed constant-width band doesn't hold.*

Weight is right-skewed in reality — the same pattern behind why weight is often log-transformed or modeled with a GLM.

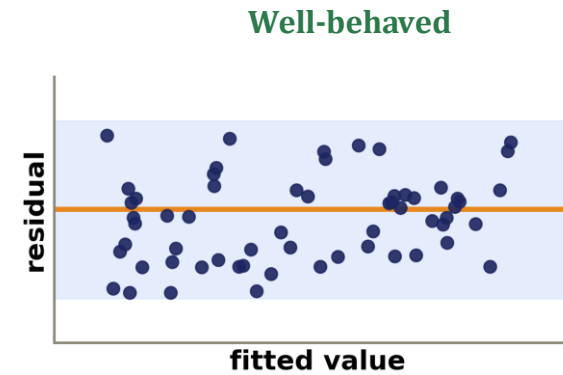
**Fixed by: GLMs (a distribution matched to the response) — covered next.**

Formal test: Shapiro-Wilk.



## E — Equal variance

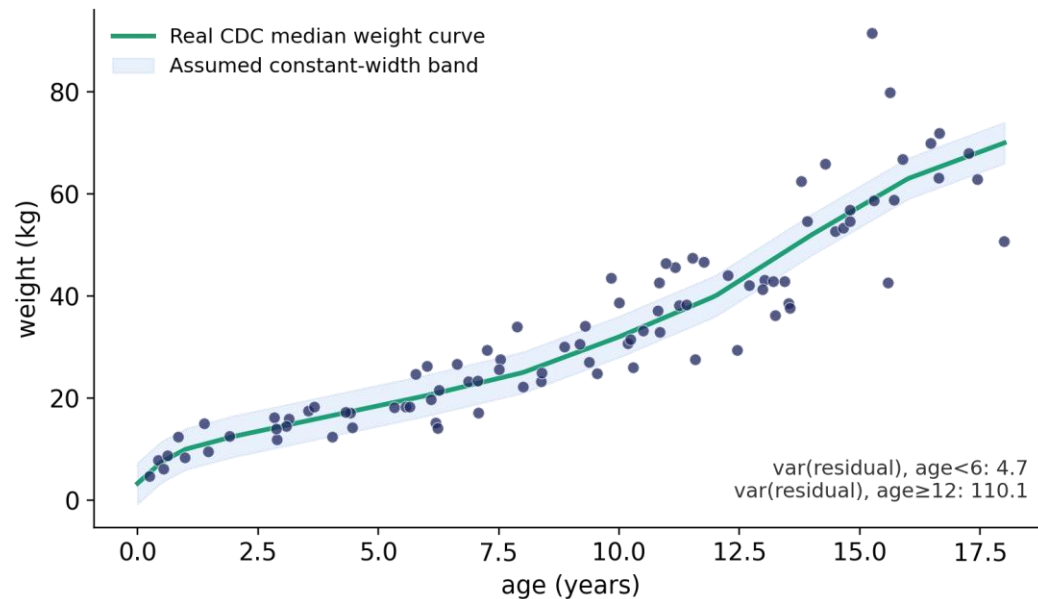
Weight residuals spread the same for toddlers and teens.



## E — Equal variance

**Weight residuals spread the same for toddlers and teens.**

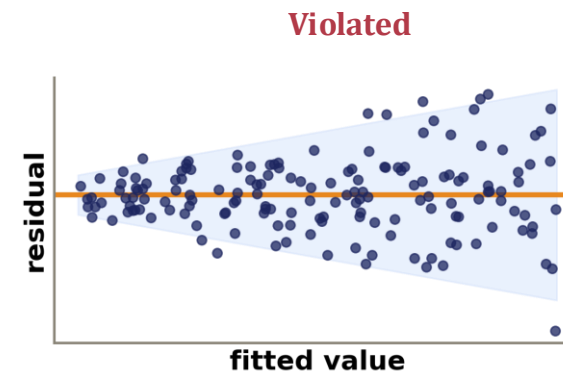
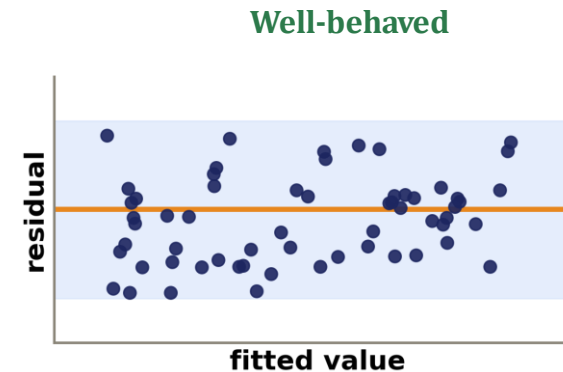
In reality residual variance is roughly 20× larger for older children than younger ones.



Same weight-vs-age data as the “N” slide — var(residual) is ~4.7 under age 6, ~110 by age 12.

**Fixed by: GLMs (variance tied to the mean, by design) — covered next.**

Formal test: Breusch-Pagan.



# What we covered today

$$Y = a + bX + \varepsilon$$

- **A model is a simplified trend**

The equation is the part we chose to describe;  $\varepsilon$  is everything we chose not to

- **One equation, however many independent variables**

Two children's equations generalize the same way — more terms, same fitting idea

- **Four assumptions of linear regression**

LINE — linear, independent, normal, equal variance — checked by looking at how  $\varepsilon$  behaves

- **Real data often breaks all four assumptions**

Growth isn't linear; children aren't independent; weight isn't equal-variance or normal

# Next: How to go about when your model breaks those assumptions ?

L

Growth isn't linear — the “L” slide

*still  $a + b_1X_1 + b_2X_2 + \varepsilon$*

**richer X (more terms, same fitting idea)**

N, E

Weight's spread and shape — the “N” and “E” slides

*linear predictor survives; link function changes what happens to it*

**GLMs**

I

Repeated measurements, same child — the “I” slide

*same fixed part, plus random effects to absorb the correlation*

**mixed effect models**

# Questions?

$$Y = a + bX + \varepsilon$$

KEEP EXPLORING

## See the code behind every plot

Every figure from this talk, rebuilt as runnable R code.

Includes a live pull of real CDC growth-chart data, not a saved file.

Change a parameter, rerun it, see what happens.



# Test yourself

8 diagnostic plots. Which LINE assumption does each one break ?

Instant feedback, running score, no login needed.





# Data sources

*All growth curves in this talk are an approximation of the data from CDC/NCHS 2000 Growth Charts. Individual data points shown alongside them are simulated, not real patient records. The companion R Notebook provided use the official CDC/NCHS 2000 Growth Charts.*

## **Landing page**

<https://www.cdc.gov/growthcharts/cdc-data-files.htm>

## **Stature-for-age, 2–20 years**

<https://www.cdc.gov/growthcharts/data/zscore/statage.csv>

## **Weight-for-age, 2–20 years**

<https://www.cdc.gov/growthcharts/data/zscore/wtage.csv>

## **Length-for-age, birth–36 months**

<https://www.cdc.gov/growthcharts/data/zscore/lenageinf.csv>

## **Weight-for-age, birth–36 months**

<https://www.cdc.gov/growthcharts/data/zscore/wtageinf.csv>

*Full citation, reproducible R code, and the knitted notebook: see the [GitHub repo README](#).*